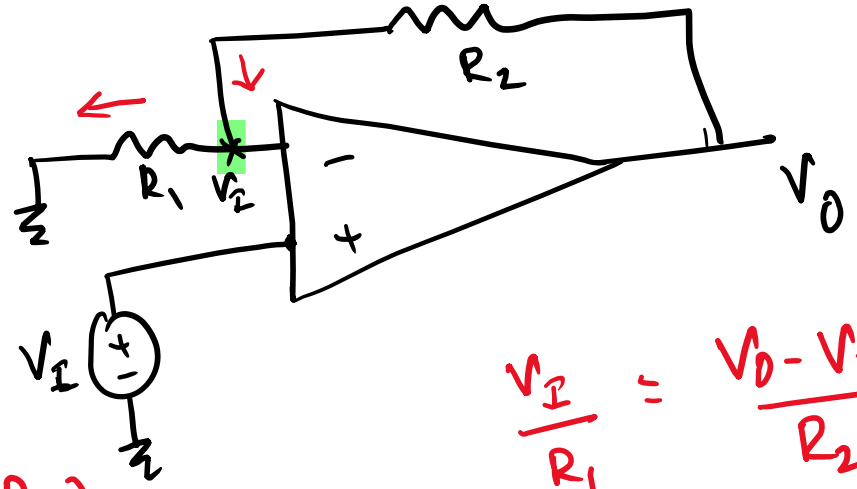
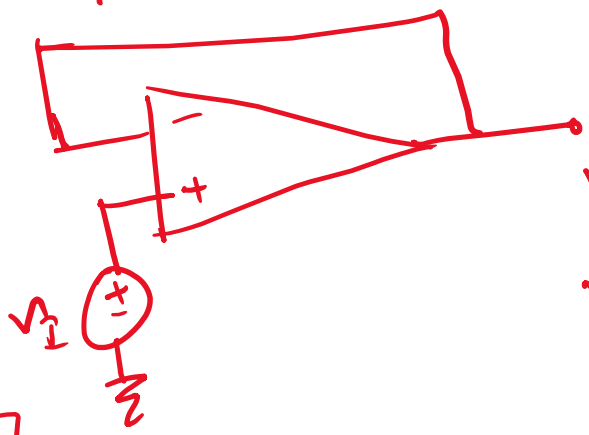
Non Inverting Configuration



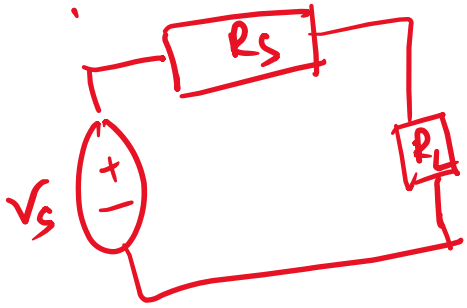
$$\frac{V_I}{R_1} = \frac{V_0 - V_I}{R_2}$$

$$V_0 = \left(1 + \frac{R_2}{R_1}\right) V_I$$

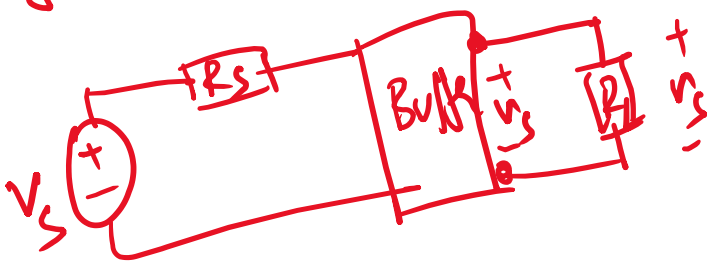
Voltage Follower / Buffer.

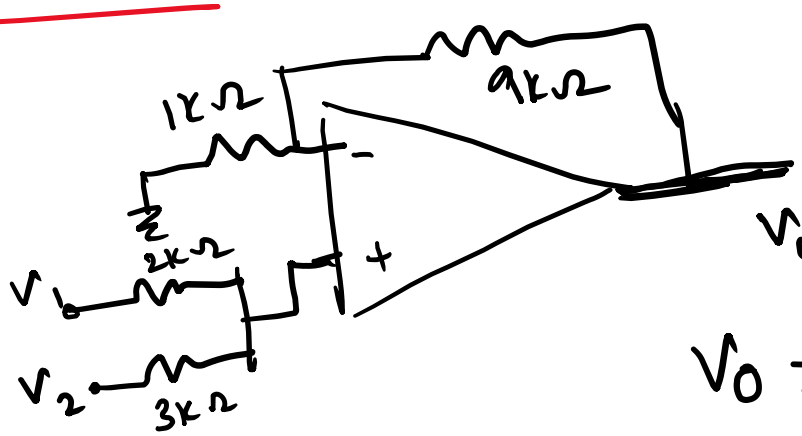


$$V_0 = V_I$$



if $R_S >> R_L$





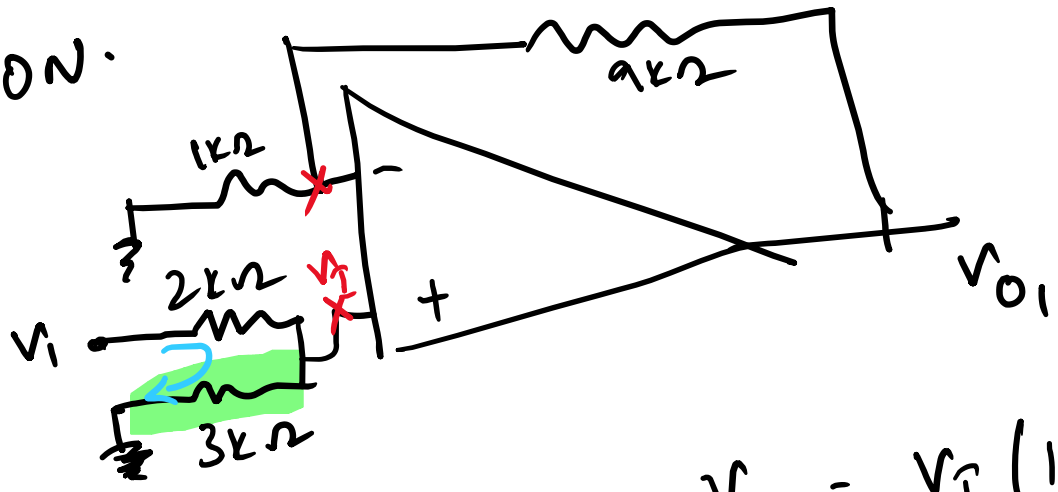
Find V_0 using superposition.

$$V_0 = V_{01} + V_{02}$$

V_{01} ↑ due to V_1

V_{02} ↑ due to V_2

V_1 is ON.

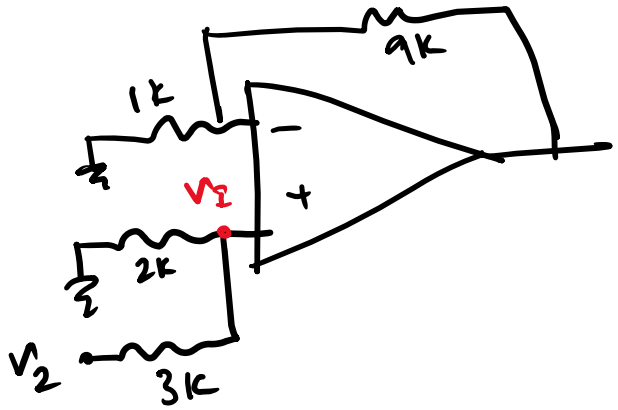


$$V_{O1} = V_I\left(1+\frac{R_2}{R_1}\right)$$

$$V_I = V_1 \cdot \frac{3k}{3k+2k} = \frac{3V_1}{5}$$

$$V_{O1} = \frac{3V_1}{5} \cdot 10$$

$$V_{O1} = 6V_1$$



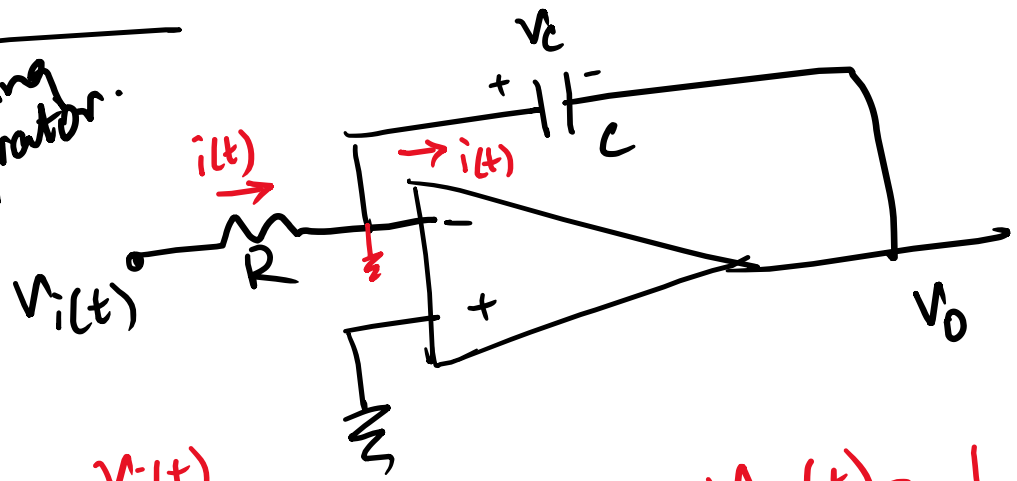
$$V_I = \frac{V_2 \times 2}{5}$$

$$V_{O2} = \frac{2V_2}{5} \times 10$$

$$= 4V_2$$

$$\boxed{V_O = 6V_1 + 4V_2}$$

Inverting Integrator.



$$V_O = -V_c$$

$$q = CV$$

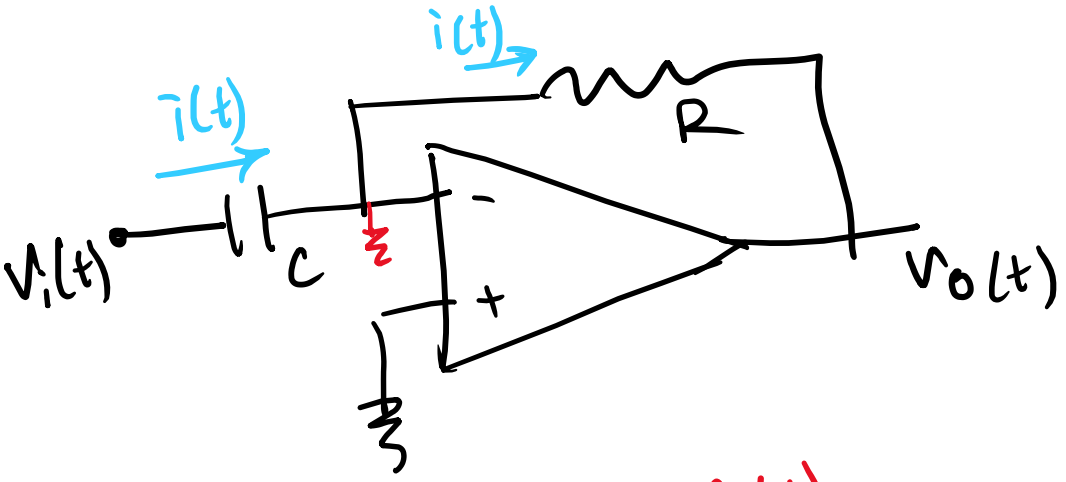
$$i = C\frac{dV}{dt}$$

$$i(t) = \frac{V_{i(t)}}{R}$$

$$V_c(t) = \frac{1}{C}\int i(t)\,dt + V_c(0)$$

$$V_c(t) = \frac{1}{RC}\int V_{i(t)}\,dt$$

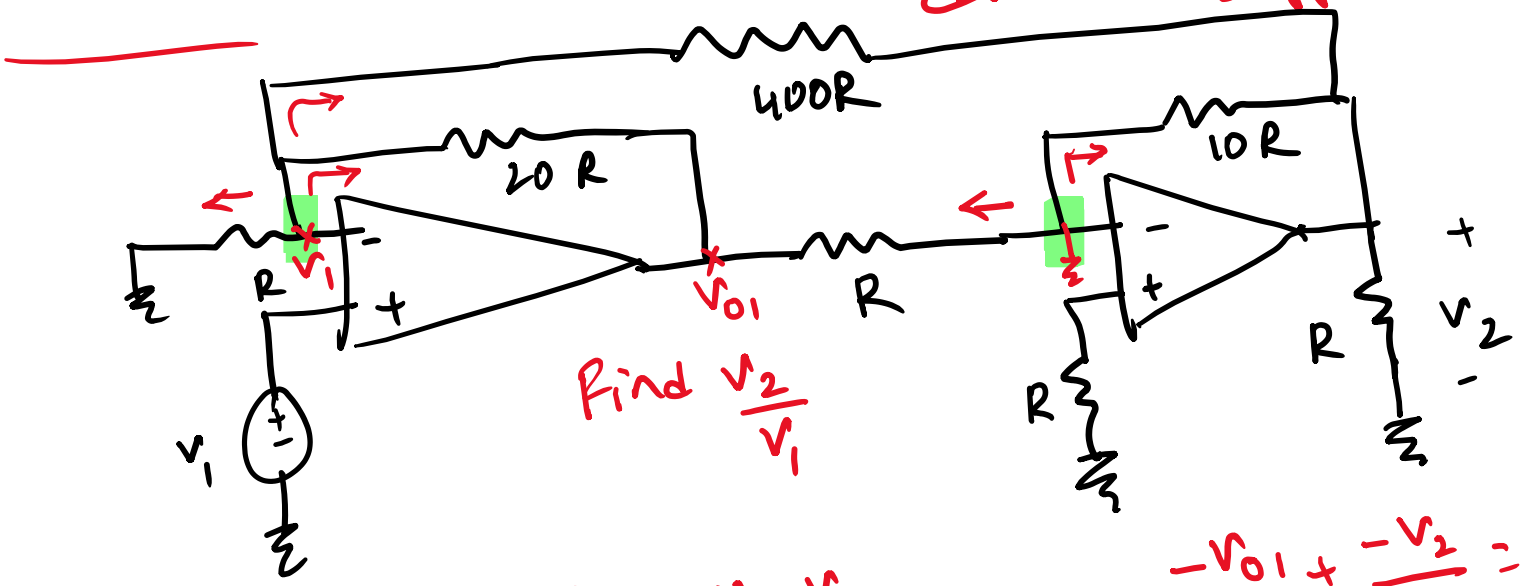
$$V_O(t) = -\frac{1}{RC}\int V_{i(t)}\,dt$$



$$i(t) = C\frac{dV_i(t)}{dt} = \frac{0 - V_o(t)}{R}$$

$$V_o(t) = -RC\frac{dV_i(t)}{dt}$$

Inverting Differentiator.

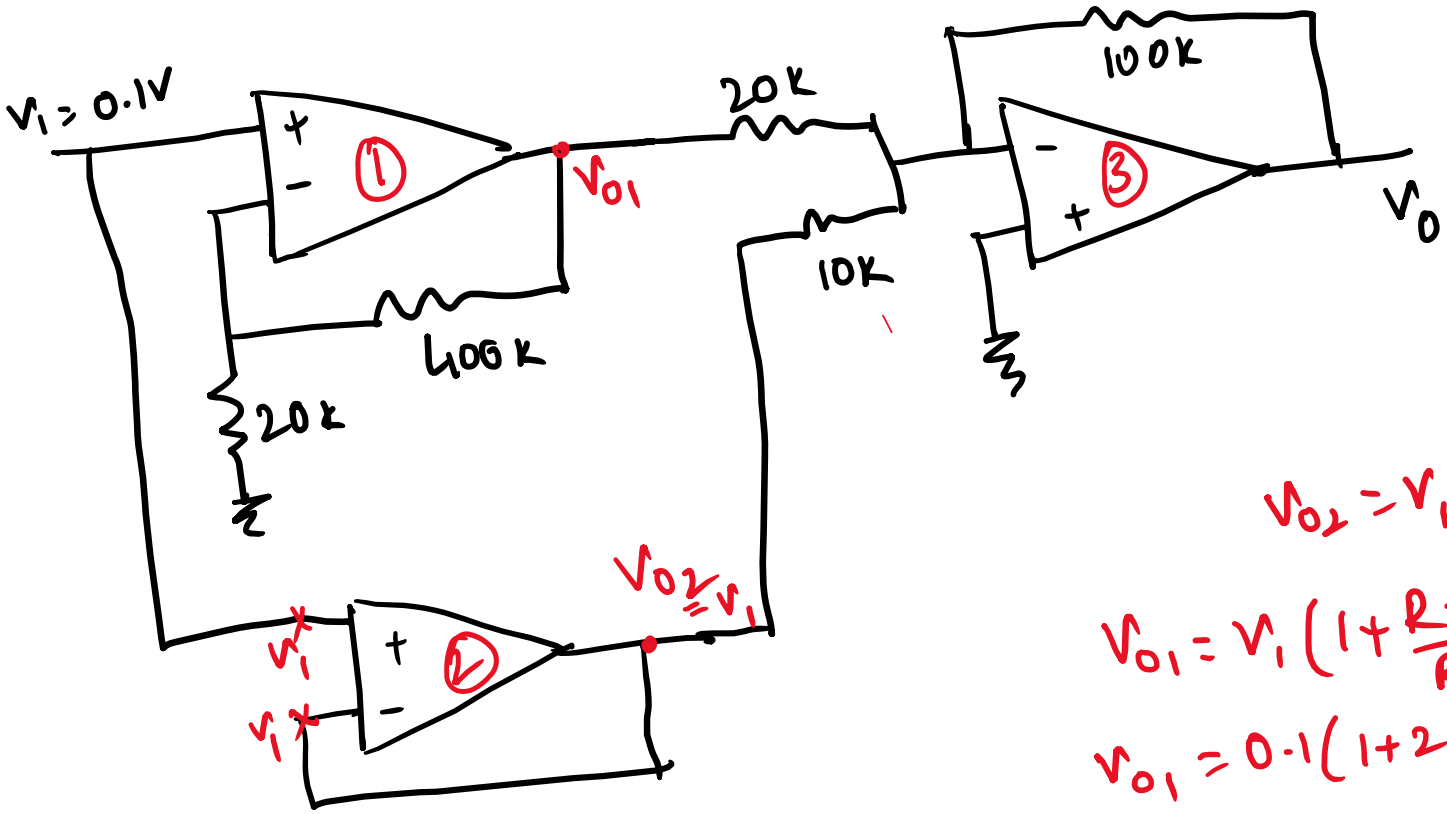


Find $\frac{V_2}{V_1}$

$$\frac{V_1}{R} + \frac{V_1 - V_{01}}{20R} + \frac{V_1 - V_2}{400R} = 0$$

$$\frac{-V_{01}}{R} + \frac{-V_2}{10R} = 0$$

$$\frac{V_2}{V_1} = -421$$



$$V_{02} = V_1$$

$$V_{01} = V_1\left(1+\frac{R_2}{R_1}\right)$$

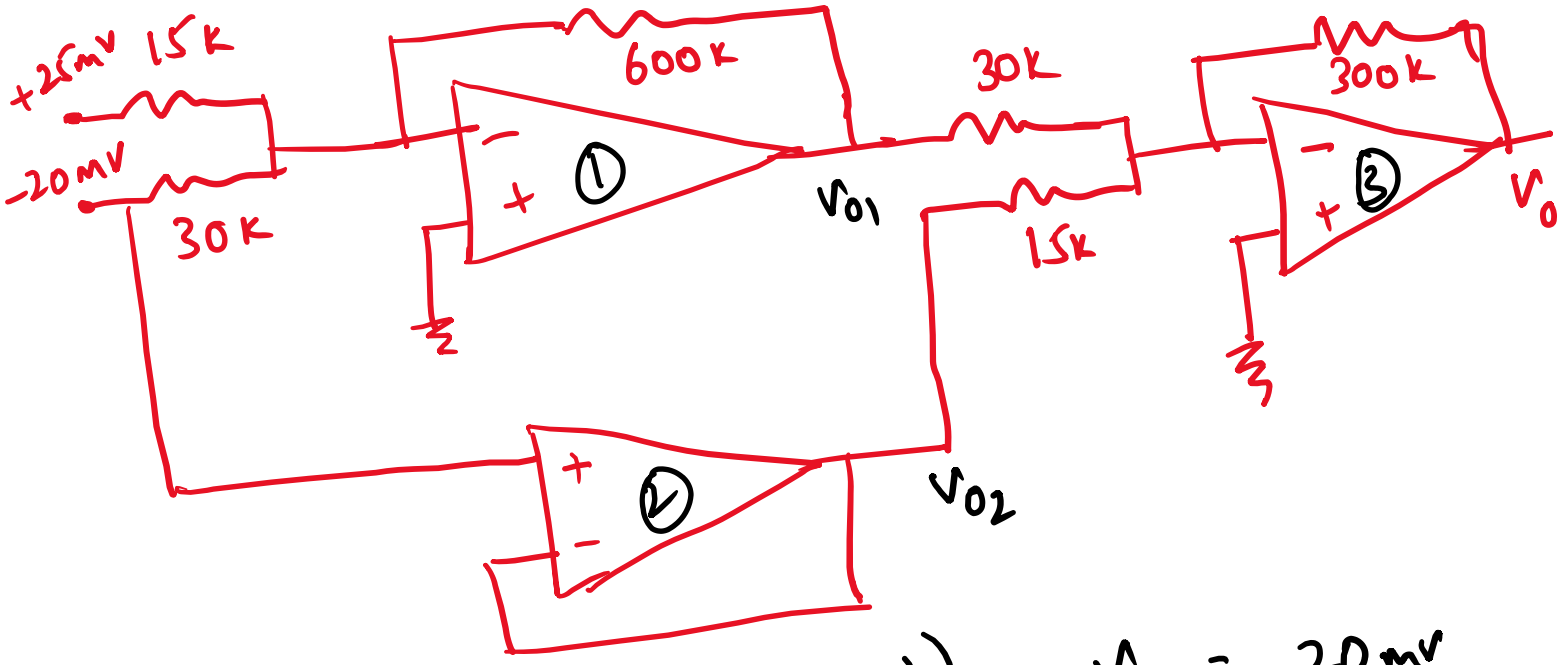
$$V_{01} = 0.1(1+20)$$

Weighted Summer:

$$V_0 = -\left(V_1 \cdot \frac{R_f}{R_1} + V_2 \cdot \frac{R_f}{R_2} + \cdots\right)$$

$$= -\left(V_{01} \cdot \frac{R_f}{R_1} + V_{02} \cdot \frac{R_f}{R_2}\right)$$

$$V_0 = -11.5V$$



$$V_{01} = -\left(\frac{600}{15}(.25mV) + \frac{600}{30}(-20)mV\right) \qquad V_{02} = -20mV$$

$$V_0 = -\left(\frac{300}{30}\cdot V_{01} + \frac{300}{15}V_{02}\right) = \underline{6.4V}$$